

Speaker notes — Growing tissues, organs & organisms

Jakub Koubele

Beyer lab retreat, 9th September 2026

How to use this. One section per slide, in deck order. Each has: what is on the screen, the spoken spine (green box), the numbers, and backup for questions. Read the green boxes and the numbers card if you are short of time; the rest is insurance. Section 13 lists things that are easy to say and wrong — worth a look on the morning.

Contents

1	The spine of this part	1
2	Slide: Oxygen supply	2
3	Slide: Self-organization of cells	4
4	Slide: Vascularization of organoids (overview)	4
5	Slide: Vascularization of organoids (strategies)	5
6	Slide: Bioprinting	7
7	Slide: Vascularization with bioprinting	9
8	Slide: Decellularized matrix	11
9	Slide: Engineered bladder	13
10	Slide: Trachea scandal	14
11	Slide: Synthetic embryos	15
12	Closing	17
13	Accuracy traps — read this on the morning	18

1 The spine of this part

Katrin's section ends by naming vascularization as the open problem of organoids. This part picks it up and does not re-introduce it.

Organoids stop at a few hundred microns because that is how far oxygen gets before cells consume it. Everything above the organoid rung is therefore a *mass-transport* problem, and there are two experimental answers: **impose the architecture** (scaffolds, bioprinting, decellularization — engineering) or **supply the right environment and let it self-organize** (chimeras, embryo models — developmental biology). Then organisms, where the bottleneck stops being oxygen and becomes the uterus.

The four slides covered here are the setup: the physical wall, why self-organization does not get past it, and what the organoid-vascularization field has actually achieved.

Numbers card

- $200\text{ }\mu\text{m}$ — how far oxygen penetrates actively consuming tissue from a supply surface. Equivalently: no cell in the body sits much further than this from a capillary.
- $300\text{ }\mu\text{m}$ – $400\text{ }\mu\text{m}$ — viable *thickness* of a slab fed from both faces (roughly twice the one-way depth). Measured directly in unsliced cortical organoids by [Qian et al. \(2020\)](#).
- **“About three sheets of paper”** — paper is $\approx 0.1\text{ mm}$. The tangible version; use it instead of the number if the room is glazing over.
- $\sim 40\%$ — fraction of a 4-month-old unsliced cortical organoid that stains hypoxic.
- $\sim 18\text{--}20\%$ vs $\sim 3\%$ — cleaved-caspase-3-positive cells in the *non-necrotic* region of unsliced vs sliced organoids.
- $500\text{ }\mu\text{m}$ — slice thickness used to fix it, chosen so nothing is more than $250\text{ }\mu\text{m}$ from a face.
- $\sim 40\times$ — how much worse culture medium is than blood as an oxygen carrier (≈ 5 vs $\approx 200\text{ mL O}_2$ per litre).
- $\sim 10\text{ }\mu\text{m}$ — capillary diameter. The target that fabrication methods keep missing.

2 Slide: Oxygen supply

On screen. Figure 1D from [Qian et al. \(2020\)](#): two unsliced human cortical organoids, day 60 and day 140, cryosectioned through the middle. Magenta is a pimonidazole hypoxia probe, white is DAPI. Scale bars $200\text{ }\mu\text{m}$.

This is a human cerebral organoid, cut through the middle. On the left it is two months old, on the right nearly five. The magenta is a hypoxia stain. On the right you can see what happens: the centre has gone — that region is necrotic — and the magenta forms a ring around it. That ring is not the damage. The stain only labels cells that are still alive and metabolically active, but starving. Inside it there is nothing left to report. About 40% of that organoid is in the starving zone, and even the outer shell that looks fine has five or six times the apoptosis of a properly oxygenated one. Nothing here can be more than a couple of hundred microns from the surface and still be doing well.

Why the ring, and not a filled disc

Two reasons stack:

- Cells are lost. Nuclei fragment and material is cleared, so there is less DAPI signal to see.
- **Pimonidazole chemistry needs living cells.** It is a 2-nitroimidazole; it only forms protein adducts after cellular nitroreductases reduce it, which requires active metabolism. A dead core physically cannot go magenta. A positive rim hugging a negative necrotic centre is the standard pattern in tumour hypoxia work.

This is my reading of the image plus how the probe works — the paper does not claim it about this panel. Their necrotic-area criteria (fragmented nuclei, loss of SOX2, diffuse caspase-3) are defined for their Figure 1F/G. If challenged: “they quantify cell death separately, with cleaved caspase-3.”

The quote worth having

“the viable thickness in unsliced organoids was restricted to 300–400 μm and **stopped increasing soon after day 70**” (Qian et al., 2020)

This is the sharpest version of the whole slide. The organoid keeps growing for another seventy days. The *living part* does not. Everything it adds after day 70, it adds dead.

The scale trap — know this before someone asks

The two panels are **not** at the same magnification. The figure legend only promises matched scale *within* a date pair. Both bars are 200 μm , but day 60 draws it with 53 pixels and day 140 with 27 — so day 140 is zoomed out about $2\times$. Measured back from the bars:

	apparent	actual
Day 60	~ 280 px	~ 1.1 mm
Day 140	~ 290 px	~ 2.1 mm

So the organoid **doubled in diameter and grew roughly eightfold in volume** between the two images, while looking the same size. Do not make a visual “look how it grew” claim — it is not visible and someone comparing bar lengths will catch it. But the fact itself is the best line available: it grew eightfold in volume and the living shell stayed at 300–400 μm .

If asked: why is the core not dead at day 60 already?

Hypoxia is not late — *necrosis* is. The organoid is already $\sim 35\%$ hypoxic at day 60. What has not happened yet is frank cell loss. Three reasons:

- **Size.** These grow to 3–4 mm over months. At day 60 the centre is still close enough to the surface to be hypoxic but survivable.
- **Demand rises.** Day 60 is progenitor-heavy; by day 100–140 there is a cortical plate full of maturing neurons, which are metabolically expensive (ion pumping dominates the neuronal energy budget). So consumption per unit volume climbs — the critical radius shrinks while the actual radius grows.
- **Hypoxia is survivable for a while.** Cells mount a HIF response and shift glycolytic. Chronic exposure over weeks is what kills.

Other backup

- **DAPI is not a viability marker.** It binds DNA in every nucleus, live or dead. In the paper it is used to outline total organoid area (the denominator for the hypoxic fraction) and to judge nuclear *morphology*. The bright dense white rings scattered through the organoid are ventricular-zone rosettes — tightly packed neural progenitors.
- **SNO** (the condition not shown) = Sliced Neocortical Organoid. Every ~ 4 weeks they slice the organoid into 500 μm slices, so no cell is more than 250 μm from a face. Crude, and it works. Note the field’s own fix independently reproduces our number.
- **Show of hands option:** “how thick can living tissue be with no blood supply?” $\rightarrow$ reveal $\rightarrow$ three sheets of paper. 20 seconds, no prep.

3 Slide: Self-organization of cells

On screen. Figure 2 from [Lancaster and Knoblich \(2014\)](#), “Principles of self-organization”. (A) cell sorting out — mixed cell types separate because of differential adhesion molecules. (B) spatially restricted lineage commitment — progenitors give rise to progeny that get displaced outward and differentiate there.

Before we go up a level, one thing worth recalling from the organoid section. The reason any of this works is that cells organize themselves. Mix two cell types and they sort out, purely because they stick to their own kind more strongly. Then fate follows position. Nobody supplies a blueprint. That is remarkable, and it is what gets you an organoid. But vessels are the exception, for two reasons. First, endothelium is a different lineage — it is mesodermal, and a brain organoid is neuroepithelium, so it simply cannot make its own. It has to be added. And second, even once you have vessels, they need flow. Vessels that never carry blood regress.

Keep this to about a minute. It is Katrin’s rung — a run-up, not a section. Bullets 1 and 2 fast; spend the time on flow.

Backup

- **The lineage point, in the review’s words:** “traditional COs lack the capacity to generate vascular endothelial cells (ECs) due to their mesodermal origins, so they must be added by other means” ([LaMontagne et al., 2022](#)).
- **Flow is instructive, not just delivery.** This is the strongest version and the thread you pick up everywhere afterwards. Shear stress from moving blood is what selects which branches of a raw vascular plexus survive and mature; unperfused branches get pruned. Hence the heart being the first functional organ in the embryo, beating well before the embryo is big enough to need transport. Evidence: Lucitti et al., *Development* 2007 (yolk-sac remodelling requires flow); Hove et al., *Nature* 2003 (intracardiac fluid forces in zebrafish).
- The figure has a panel B (lineage commitment) that the bullets do not mention. If you want to cover it: “and fate follows position.”

4 Slide: Vascularization of organoids (overview)

On screen. Figure 2 from [LaMontagne et al. \(2022\)](#). A hub — “Vascularized Cortical Organoid”, with O₂, glucose and nutrients flowing in and CO₂ and metabolic waste flowing out — surrounded by four quadrants: Cellular Signaling, Hemodynamic Forces, Extracellular Matrix, Engraftment *In Vivo*.

So what would it actually take. Start in the middle: this is what a blood supply is for. Oxygen and nutrients in, carbon dioxide and waste out — and it is worth remembering the second half, because we tend to talk about this as an oxygen problem when clearing waste is just as much of a constraint. Around the outside are the things you need to get there. Top left is the signalling, and I am not going to read those out — the point is just that it is complicated, and it is cross-talk between the neural cells and the vascular cells, not either one alone. Bottom left, matrix. And then the two that matter for the rest of this talk: you need flow, and flow has to come from somewhere. Either you supply it with a machine — top right, a microfluidic chip — or you put the thing inside an animal and let the animal do it. Hold that mouse in the corner; we come back to it.

The gene names (TGF- β 1, Wnt7a/7b, VEGF, SLIT2/ROBO4, ANG1/2) function as a *signifier of complexity*, not as content. Do not read them out.

If asked: organoid-on-chip — is it blood?

No. It is **culture medium**, driven by a syringe or peristaltic pump (sometimes gravity or a rocker). Essentially nobody perfuses organoids with real blood, because:

- **It clots.** Blood needs a mature endothelium expressing anticoagulant surface factors to stay liquid. Bare hydrogel channels, or immature endothelium, and you get a clot within minutes.
- **Pumps destroy blood cells** — shear-induced haemolysis.
- It is undefined, variable, immunologically active, and would need a continuous donor supply.

The cost of using medium is a good number. Blood carries ≈ 200 mL O₂ per litre thanks to haemoglobin. Medium or plasma dissolves only ≈ 5 mL per litre. Medium is roughly a **40 $\times$ worse oxygen carrier**. That is exactly why Skylar-Scott et al. (*Sci. Adv.* 2019) perfused with 95% O₂ — it compensates for medium being a poor carrier. Some groups add haemoglobin-based carriers or perfluorocarbons for the same reason.

Filtration: essentially none. Systems are single-pass (medium out to a waste reservoir) or recirculating through a reservoir. The “kidney” is somebody changing the medium. Multi-organ “body-on-a-chip” work does try to link a liver module for metabolism and sometimes a kidney module for clearance, but that is a research goal, not standard practice.

A chip gives you flow, not blood. It solves shear and convection. It does not solve oxygen-carrying capacity or waste clearance.

5 Slide: Vascularization of organoids (strategies)

On screen. Panels 3A, 3B and 5A from [LaMontagne et al. \(2022\)](#): endothelial cell addition, vascular spheroid fusion, and engraftment into a mouse brain.

So here is what people have actually tried. First, just mix endothelial cells in with the stem cells at the start and hope they organize — that works, in the sense that you get vessel-like structures, and it is currently the best of the in-vitro methods. Second, grow the vasculature separately as its own spheroid, stick the two together, and let the vessels invade. There is a third I am not showing, which is to genetically induce the organoid to make its own endothelium from within — switch on a transcription factor called ETV2 and some of the cells become vascular. All three produce vessels. None of them produce a blood supply. The only thing on this slide that gives you real perfusion is the one on the right: put the organoid into a living mouse and let the mouse grow into it. You can prove it — inject a labelled dye into the mouse’s bloodstream and it turns up inside the human graft.

The strategies, with sources

- **3A — Endothelial cell addition.** HUVECs co-cultured with PSCs from the start. Shi et al., 2020. Crudest, and per the review the most successful *in vitro* route.
- **3B — Spheroid fusion.** Grow a vascular organoid separately, fuse, let vessels invade the neuroepithelium. Wörsdörfer et al., 2019.

- **3C — Co-differentiation** (not on the slide). Either soluble factors (VEGF, ANG, TGF- β , WNT7) or genetic: doxycycline-inducible **ETV2** turns a subset of the organoid’s own cells into endothelium. No foreign cells at all. Cakir et al., 2019; Ham et al., 2020.
- **5A — Engraftment.** Human organoid into adult mouse brain. Host vessels invade and *perfuse*. Mansour et al., 2018; Shi et al., 2020. Their Figure 5h: dextran injected into the host mouse appears in vessels inside the graft.

The arc across the first three is worth naming out loud: add cells $\rightarrow$ add tissue $\rightarrow$ make the organoid grow its own. Increasingly elegant, and none of them solve it.

The verdict, in the review’s words

“Of these categories, **engrafted organoids in vivo are most successful**. They vascularized robustly and show strongest evidence of vessel maturity and perfusability, but because they rely on **non-human hosts**, COs vascularized in vivo are rendered imperfect representations of human neurovasculature and unsuitable for modeling human disease or development.”

“**none** fully capture the form and function of an in vivo neurovascular unit.”

The primary objection to engraftment is that the vessels are mouse vessels — mouse endothelium, mouse blood-brain barrier — which defeats the point if you wanted a human model. Throughput, cost, months of waiting, graft-to-graft variability, the size ceiling of a mouse brain, and the ethics of human neural tissue in animal brains are all genuine, but secondary.

Two more quotable lines (both p. 11)

“TE vessels currently fall short in producing realistic vasculature, often generating **vessels adjacent to organoids rather than through them**.”

The failure in one phrase: the vessel and the tissue end up next to each other, not integrated.

“TE vessels are usually **significantly larger than cortical microvessels and lack branches** due to limitations by their fabrication methods.”

This is the resolution problem, stated early. It sets up the bioprinting section: print resolution versus $\sim 10\ \mu\text{m}$ capillaries.

“The best of these being EC addition during organoid formation, which results in dense and highly branched networks that span the majority of the organoid... Yet, biologically-based self-organization approaches may be **incapable** of accomplishing the same level of vessel interconnectedness as models that incorporate flow, since the development of this feature **relies on fluid forces**.”

That closes the loop with the self-organization slide: self-organization gets you a long way and then stops, because the missing ingredient is physical.

If asked: does spheroid fusion scale to bigger tissue?

No, and the reasons are the bridge to the next section.

- **It never establishes flow.** Fusion adds endothelial *cells*, not a supply. Nothing is pumping. So you have made a bigger avascular object, which is strictly worse. The wall does not move.

- **Timescales do not match.** Sprouting angiogenesis advances on the order of tens of microns per day. Necrosis sets in within hours to days — Skylar-Scott et al. saw a necrotic core in 12 hours. Crossing a centimetre by sprouting would take most of a year.
- **Surface-to-volume.** Fusion is a *surface* operation: vessels enter across the interface. Interface area scales as r^2 , volume as r^3 . At organoid scale the interface is a large fraction of the object; at organ scale it is a rounding error.
- **No inlet, no outlet.** Even if a plexus formed throughout, there is no artery to cannulate and no vein to drain. You cannot connect it to anything.

That last point is the pivot. The problem is not “how do we get endothelial cells in there”. It is “how do we get a plumbed tree with a connector on it” — one large vessel in, branching down to capillaries, recollecting on the other side. There are exactly two answers, and they are the next two sections: print the tree, or steal one from an organ that already has it.

Fair caveat if pushed: spheroid fusion is not a failed idea. **Assembloids** are thriving for *modelling* — fusing cortical and striatal organoids to study circuits between regions. It is just not a route to vascularizing something organ-sized.

6 Slide: Bioprinting

On screen. Figure 1 from [Idnani and Khan \(2026\)](#): the three-step workflow. **Preprocessing** — medical imaging (CT/MRI) converted to a 3D CAD model. **Processing** — bioink deposited layer by layer. **Postprocessing** — the construct matured in a bioreactor.

Slide bullets are deliberately number-free; the numbers below are for saying aloud.

The other way to get architecture is to stop waiting for the cells to arrange themselves and just put them where you want them. You scan a real organ, you build a CAD model from the scan, and you print it layer by layer out of a bioink — which is cells, plus a material to hold them, plus signalling molecules. Then it goes into a bioreactor to mature, because printing it is not the same as finishing it. Two things make this hard. You need an enormous number of cells, and they have to already be the cell type you want. And every printing method forces you to choose between resolution, how thick and cell-dense your ink can be, how fast you go, and how many cells survive the nozzle. Nobody has all of them at once.

Bioink: what is actually in it

The review’s definition: bioinks are “composed of **cells, biomaterials and signaling molecules**”. Note that the “biomaterial” is usually *not* real ECM — common choices are alginate (a seaweed polysaccharide), Pluronic (a surfactant), GelMA (chemically modified gelatin) and PEG.

The matrix has two jobs that pull against each other:

- **Print** — flow through a nozzle under shear, then hold shape immediately. A rheological requirement.
- **Support cells** — adhesion motifs, right stiffness, degradable on the right timescale.

Materials good at one are bad at the other. Alginate prints beautifully and cells cannot adhere to it at all (no integrin-binding sites). Collagen is biologically ideal and prints terribly — which is exactly why **FRESH** exists: extrude collagen into a supporting gel bath so it has time to gel instead of collapsing. If you show the FRESH video, that is what you are looking at: a workaround for this conflict.

Frontier, and the bridge to the next section: dECM bioinks — decellularized ECM from real tissue, ground up and made printable. It keeps tissue-specific structural proteins and glycosaminoglycans, and it is *instructive*: stem cells printed in cartilage dECM undergo chondrogenesis, in adipose dECM adipogenesis. Same material as the decell section, used two ways — kept intact as a whole scaffold with its vascular tree, or ground up as ink.

Resolution by modality — the numbers

Modality	Resolution	The catch
Extrusion	150–350 μm filaments	the only one handling cell-dense inks
Inkjet	10–50 μm droplets	needs dilute inks, 3–30 mPa s
SLA	~ 100 μm feature size	photosensitive materials, UV crosslinking
DLP / two-photon	10–50 μm	penetration depth also only 10–50 μm
Multi-material rig	25 μm	up to six bioinks simultaneously

Capillaries are ~ 10 μm . Printing speeds: 10–50 $\mu\text{m/s}$ for high resolution, 1–30 mm/s for lower-resolution volumetric printing. Shear stress above 1–5 kPa costs **40–80% cell viability**, so pushing for resolution kills cells.

Nobody gets all three at once. Fine resolution comes with dilute inks, or very thin layers, or photochemistry. Cell density and thickness come with coarse filaments. The methods that reach capillary scale cannot carry the cells; the method that carries the cells cannot reach capillary scale.

The detail that makes it land: **two-photon polymerization reaches 10–50 μm resolution, but its penetration depth is also 10–50 μm .** The one technique with genuine capillary resolution only works within a layer thinner than the problem it is meant to solve. That is why the trade-off is structural rather than a matter of better engineering.

Why the cells are the harder problem

Bioprinting needs *differentiated* cells, and this follows from the axis: organoids need cells that can still decide (the differentiation *is* the mechanism); printing needs cells that will stay where they are put. The source is usually still pluripotent — iPSC-derived cardiomyocytes, iPSC-derived hepatocytes — you just differentiate *before* printing rather than during. There is also a safety reason: residual undifferentiated cells in a graft are an immunological and tumour risk.

The cost is that differentiated cells barely proliferate, so a printed organ must be handed its entire cell count on day one, whereas an organoid grows its own from a few thousand starting cells. Hence:

Adult liver	3.6×10^{11} cells (70% hepatocytes)
Human heart	$2\text{--}3 \times 10^9$ cardiomyocytes ($< 1/3$ of cardiac cells)
Best PSC bioreactor batch	$\sim 10^{10}$ cells
Differentiation at $>90\%$ purity	$10^8\text{--}10^9$ scale

So a liver requires pooling many of the largest runs anyone can do, with batch-to-batch variation in purity accumulating across them. The review’s verdict: graft-level (10^9) is practical now; whole-organ (10^{11}) “remains a key frontier”.

Organoids are cheap in cells and expensive in control. Printing is cheap in control and expensive in cells.

That is why **SWIFT** (Skylar-Scott et al., *Sci. Adv.* 2019) is the clever move: print *with organoids as the ink*. Self-organization makes the cells inside each brick; printing imposes the architecture

between bricks. Their matrix runs at ~ 200 million cells/ml — real tissue density — from $\sim 400,000$ embryoid bodies, half a billion cells in 2.5 ml. Figure 1E of that paper shows they did it with EBs, cerebral organoids and cardiac spheroids.

7 Slide: Vascularization with bioprinting

On screen. Figure 5 from [Idnani and Khan \(2026\)](#), four panels: (1) sacrificial templating, (2) coaxial extrusion, (3) direct endothelial cell printing, (4) the final perfusable construct on a pump. Image only — everything below is spoken.

So how do you actually print a blood vessel? The awkward part is that a vessel is a hole, and you cannot print a hole — if you try to extrude a hollow tube out of soft gel it collapses before it sets. There are two tricks. The first, top left, is to print the vessel as a *solid* strand of a material you intend to throw away, build the tissue around it, and then dissolve the strand out. What is left behind is the channel. It is lost-wax casting. The second, top middle, is a nozzle inside a nozzle: you push throwaway material through the middle and cell-laden gel through the outside, and a hollow tube comes out in one go. Then, top right, you can also just print endothelial cells into the bulk and let them find each other. And the bottom panel is the point of all of it: a block of tissue with an inlet, an outlet, and a pump — which is exactly the thing that was missing two slides ago.

The core problem: you cannot print a hole

Everything in this figure is a workaround for one fact. A blood vessel is defined by its lumen — by absence. An extruder deposits material; it cannot deposit absence. And an unsupported hollow tube extruded in a soft hydrogel collapses under its own weight before the gel crosslinks. So the lumen has to be created indirectly.

1. Sacrificial templating — print the negative

The procedure:

1. Print a **solid** filament of *sacrificial ink* tracing the path of the vessel network you want. Branched geometry is fine — you are just drawing a shape.
2. Cast or print the **cell-laden matrix** around it.
3. Crosslink the surrounding matrix so it holds its shape on its own.
4. Remove the sacrificial material.
5. The void it leaves **is** the channel.

Analogy for the talk: lost-wax casting. Model the object in wax, encase it in plaster, melt the wax out, and the cavity is your object.

Two inks, and only one has cells in it. This is the part that is easy to get backwards. The sacrificial ink is *acellular* — it is a pure placeholder. The cells are in the bulk matrix, which stays. Nothing living is ever washed away.

Why these particular materials. A sacrificial ink has to vanish under conditions the surrounding cells survive, print well at room temperature, and not crosslink into the matrix around it. That is a narrow window, hence a short list:

Material	How it is removed
Pluronic F-127	reverse-thermal gel: solid when warm, <i>liquid at 4 °C</i>
Gelatin	melts at 37 °C
Carbohydrate glass	dissolves in water/medium

The review’s point about this: the process “avoids harsh solvents or extreme temperatures, maintaining the viability of surrounding cells”.

Consequence for resolution. The channel is a *cast* of the strand, so channel diameter = filament diameter. That is the direct link to the resolution numbers on the previous slide, and it is why sacrificial channels come out at **1–5 mm**.

You already have an example: this is the SWIFT method — sacrificial ink printed into the compacted organoid matrix, evacuated, then perfused.

2. Coaxial extrusion — the tube arrives hollow

A **coaxial nozzle** is a nozzle inside a nozzle, extruding two streams at once:

- **Core** (inner stream): sacrificial material, or a crosslinking agent
- **Shell** (outer stream): cell-laden hydrogel

The shell gels as it leaves the nozzle — frequently crosslinked *from the inside* by the core (e.g. calcium in the core setting an alginate shell). A hollow tube emerges in a single step, walls already formed, lumen already open, ready to be lined.

Analogy: extruding a drinking straw, or toothpaste from a tube-within-a-tube.

Which to use — and why the paper splits them this way

	Steps	Good for
Sacrificial	two: print solid, then dissolve	branched networks — any path you can draw
Coaxial	one: hollow on exit	individual vessels — proper wall, but hard to branch

Coaxial lays continuous hose, so branching is awkward; sacrificial is just drawing, so a hierarchical tree is straightforward. That is exactly how their Table 2 assigns them: sacrificial to “hierarchical vascular networks”, coaxial to “blood vessels from capillaries to arteries”.

Do not confuse this with embedded printing

Printing temporary material to hold soft structures up *during* printing is a different technique — **embedded bioprinting**, its own row in their Table 2 (“printing within a support to stabilize soft hydrogels”). **FRESH** is this: extrude collagen into a support bath so it gels before it slumps. The two are often combined — sacrificial ink printed *inside* a support bath — but they solve different problems. Sacrificial makes a void; embedded stops a slump.

3 and 4. Endothelial cells and the pump

Panel 3, direct EC printing: put endothelial cells in the bioink and let them self-assemble into lumens. This is self-organization used *inside* the impose-the-architecture approach — worth naming out loud, since it is the theme of the talk showing up within a single construct.

Panel 4, the final construct. Two things happen here.

Endothelialization is a separate operation after the channel exists. Perfuse a suspension of endothelial cells through the empty channel — SWIFT used HUVECs at 10^7 cells/ml — let them settle onto the lumen wall, and keep perfusing. Over ~20 hours they spread into a confluent monolayer with VE-cadherin at the cell-cell junctions. A real endothelium, not just cells stuck to gel. (SWIFT Figure 3E–H.)

Why the lining matters: without it you have a pipe through jelly. The endothelium provides barrier function, an anticoagulant luminal surface (necessary if real blood ever goes through), and shear sensing — it is what responds to flow and remodels. It is the difference between a channel and a vessel.

Perfusion is a peristaltic pump driving medium, not blood — see the 40× oxygen-carrying-capacity point in the previous section.

The verdict: plumbing, not capillaries

Bioprinting solves the plumbing. It does not solve the capillaries.

This is the payoff of the “no inlet, no outlet” pivot: a printed construct has a cannulatable inlet, an outlet, and a pump. That is precisely what spheroid fusion could not give you. But the channels are **1–5 mm** and capillaries are $\sim 10\ \mu\text{m}$. Their own Table 2 lists resolution or accuracy as the limitation of every single strategy:

Strategy	Limitation (their words)
Coordinated patterning	low scalability; fidelity low in tissues $>5\ \text{mm}$
Sacrificial bioinks	low printing accuracy
Embedded bioprinting	low mechanical integrity
Coaxial	low resolution & microenvironment complexity

The review does claim this prevents hypoxia in tissues “exceeding 1 cm in thickness”. Treat that with mild suspicion, for the reason below.

Why you cannot just print bigger channels closer together

Back-of-envelope, so present it as order-of-magnitude — but it holds, and it is the strongest argument for why capillary scale is not a fussy detail.

Tissue must be within $\sim 200\ \mu\text{m}$ of a supply. So a **1 mm** channel needs its neighbours about $400\ \mu\text{m}$ away, meaning each channel plus the tissue it feeds is $\sim 1.4\ \text{mm}$ across, of which 1 mm is hole. **Over half the construct is lumen.** That is not tissue, it is a sponge.

A real capillary bed: $10\ \mu\text{m}$ vessels roughly $200\ \mu\text{m}$ apart. Lumen fraction well under 1%.

Capillary scale is the only scale at which a vascular network can supply tissue without displacing it. Bigger channels cannot be packed densely enough to substitute, no matter how many you print.

Where the field actually hopes to go

Combine panels 1 and 3: print the large channels, *and* put endothelial cells in the bulk matrix, so they sprout outward from the printed channels and meet in the middle — growing the fine end of the hierarchy biologically because it cannot be printed. (Same angiogenic-sprouting idea as LaMontagne’s Figure 4B.)

Print the trunk, grow the twigs.

And that is the natural setup for the next section: a decellularized organ hands you a tree that already goes all the way down to capillaries, because something else built it.

8 Slide: Decellularized matrix

On screen. Figures 1c and 4d from [Ott et al. \(2008\)](#) — the SDS decellularization photo sequence, and the pacing traces.

The other way to get a scaffold is to not build one at all. Take a heart, and perfuse detergent down the aorta for twelve hours. The cells wash out and what is left is the extracellular matrix — the shape of the heart, in white. And crucially, the entire vascular tree survives it: they filled the coronaries with resin and dissolved the tissue away, and the cast is identical to a normal heart's, right down to third- and fourth-order branches. Nobody printed that. Then they put rat cells back in, kept it in a bioreactor with a pump and a pacemaker, and after eight days it beat — at about two percent of the force of a real rat heart. That was 2008. There is still no working decellularized organ in a person.

What the procedure is

Cadaveric adult **rat** hearts mounted on a Langendorff apparatus; retrograde perfusion down the aorta for 12 h. They compared three detergents: PEG and Triton X-100 left nuclei and myofibers behind (incomplete); **SDS** removed everything. If asked, that is one sentence — “it took some work to find a detergent that removes the cells without wrecking the matrix.”

The vascular evidence you are not showing but should know

- **Figure 3a** — coronary corrosion casts (resin injected into the vasculature, tissue then dissolved away) of a normal versus a decellularized heart. Identical. Their text: “both the larger cardiac vessels and the smaller **third- and fourth-level branches were patent.**” The cast is heart-shaped because the vessels go everywhere.
- **Figure 3b** — the decellularized heart transplanted heterotopically into a living rat, photographed before and just after the host aorta is unclamped: pale ghost becomes a network of bright red vessels. Note it is *heterotopic* (plumbed in elsewhere, not replacing the animal's heart) and the scaffold is still *acellular*. It is a plumbing test, not an organ.

Recellularization — and why it is the hard half

No human cells anywhere in this paper. They used **neonatal rat cardiac cells** and **rat aortic endothelial cells**. “Human” appears only as a yardstick (25% of a 16-week fetal human heart).

The two cell types went in by *different routes*, and this is the illuminating part:

- **Endothelial cells: perfusion seeding.** Flowed in through the vasculature. After 7 d: 550.7 ± 99.0 cells/mm² on the endocardial surface, 264.8 ± 49.2 within the wall.
- **Cardiac cells: intramural injection.** Injected into the muscle wall with a needle, $n = 8$ hearts.

You can deliver endothelium through the plumbing, because lining the plumbing is what endothelium does. You cannot deliver cardiomyocytes that way — they belong in the muscle wall, on the far side of a vessel wall that has no endothelium and no capillaries small enough to reach them. So you inject, and hope.

Hence the number: recellularization was **$33.8 \pm 3.4\%$** per cross-section near the injection sites (within 1–2 mm) and “decreased distally”. The cells only went where the needle went.

Function, honestly

Working-heart bioreactor: left atrium and ascending aorta cannulated, oxygenated medium in through the atrium and out through the aortic valve, physiological preload and afterload, electrical pacing at 5–20 V, maintained 8–28 days. By day 8 the constructs depolarized and contracted to

pacing, generating ~ 2.4 mmHg — **$\sim 2\%$ of adult rat heart function** (and 25% of a 16-week fetal human heart). Three of eight preparations were stopped early because of infection.

Also: the preserved vessels come **bare**. Figure 3d — “endothelial cells or nuclei were absent.” You inherit the pipes with no lining.

If asked: is this a clinical path?

The pitch is real and it is what the field envisions: take a donor heart unsuitable for transplant (or a pig heart), decellularize it, reseed with the *recipient's own* iPSC-derived cardiomyocytes and endothelial cells, mature it in a bioreactor, transplant with no immunosuppression. It would solve rejection, architecture and donor supply at once.

Why it has not happened in eighteen years:

- **Cell number.** A human heart is 2–3 billion cardiomyocytes, ~ 10 billion cells total. Best bioreactors: $\sim 10^{10}$ per batch. Same wall as bioprinting.
- **Delivery.** See above — nobody can distribute two billion cells evenly through a 300 g organ.
- **Thrombosis — the immediate showstopper.** Bare decellularized collagen is among the most thrombogenic surfaces in biology; it is exactly what the clotting cascade evolved to detect. Connect a partially endothelialized scaffold to a circulation and it clots within minutes. You need essentially *complete* endothelial coverage of the whole tree first.
- **Maturation.** iPSC-derived cardiomyocytes are fetal-like and arrhythmogenic — the same fetal-identity problem as organoids.
- **Time.** Deriving and differentiating a patient's own iPSCs takes months. Hence the pivot to off-the-shelf hypoimmunogenic lines.

What is in the clinic: decellularized heart valves, human dermis (AlloDerm), porcine small intestinal submucosa, nerve grafts, vascular grafts — all routine. They go in **acellular** and the patient's own cells migrate in afterwards. That works for thin, mechanically simple tissue. A heart needs two billion working cells on day one.

Decellularization did not solve the problem, it moved it. Bioprinting gives you control over the architecture and cannot reach capillary scale. Decellularization hands you a perfect capillary architecture and no way to fill it.

9 Slide: Engineered bladder

On screen. Figure 1 from [Atala et al. \(2006\)](#): (A) scaffold seeded with cells, (B) engineered bladder sutured to the native bladder, (C) implant covered with fibrin glue and omentum.

It is not all failure. In 2006 Atala's group did this in seven patients — children and young adults with bladder disease. They took a biopsy from the patient's own bladder, grew the cells up, seeded them onto a collagen scaffold, and sewed it in. And then they did the thing that made it work: they wrapped it in omentum, which is a sheet of fatty, very well-supplied tissue from the abdomen. The host's own blood vessels grew into the graft. So Atala did not solve the vascularization problem. He outsourced it to the patient. And it worked, because a bladder is about the friendliest case there is — thin, hollow, and it does almost nothing metabolically. It never gets anywhere near the wall we started with.

Details

- Seven patients with myelomeningocele, aged 4–19 (mean 11).
- Autologous urothelial and muscle cells from the patient’s own bladder biopsy, expanded and seeded onto a collagen–PGA composite scaffold.
- **Wrapped in omentum** after implantation. The paper’s own conclusion sentence names it, and they note the wrap “improved tissue vascularisation”.
- Follow-up 22–61 months (mean 46).
- Their Figure 3: biopsies at 31 months show a proper tri-layered structure — urothelium, submucosa, smooth muscle.

The pattern worth naming

Tissue-engineering’s clinical wins are all **flat, thin or hollow**: skin, cornea, bladder, urethra, blood vessel. The failures are all **solid and dense**: heart, liver, kidney. That is not a coincidence about difficulty — it is the diffusion limit from slide 1 showing up as clinical outcomes.

10 Slide: Trachea scandal

On screen. A retracted Lancet paper (RETRACTED watermark) and BBC coverage.

And here is what happens when you take the same idea to patients before the biology is ready. Between 2008 and 2013 a surgeon called Paolo Macchiarini implanted engineered tracheas — first decellularized donor tracheas, later ones made of plastic — seeded with the patients’ own cells. It was reported as the first tissue-engineered organ transplant. Most of the recipients died. Four Lancet papers have been retracted, and in 2023 a Swedish appeals court convicted him of gross assault and sent him to prison for two and a half years. But here is the part that matters for this talk: even without any of the misconduct, it was never going to work — because a trachea has no single artery. Its blood arrives through dozens of tiny vessels along its length, and there is nothing a surgeon can connect. The graft went in with no blood supply at all.

The two papers — do not conflate them

- **2008, *Lancet*** — Macchiarini et al., “Clinical transplantation of a tissue-engineered airway”. A *decellularized donor* trachea seeded with the patient’s own epithelial cells and MSC-derived chondrocytes, in Barcelona. This is the one that connects to the decell slide. **Retracted October 2023**, along with its 2013 five-year follow-up.
- **2011, *Lancet*** — Jungebluth, Macchiarini et al., “Tracheobronchial transplantation with a stem-cell-seeded bioartificial nanocomposite”. A *synthetic* plastic (POSS-PCU) scaffold seeded with bone marrow cells, at Karolinska. **Retracted 2018**. Around eight patients received synthetic tracheas; most died.

Timeline

2014, four Karolinska clinicians file misconduct complaints, initially dismissed → 2016, the SVT documentary *Experimenten* airs; the Karolinska vice-chancellor and members of the Nobel Assembly resign → 2018, Karolinska finds misconduct (no ethical permit, insufficient preclinical data, data presented “unduly positive and uncritical”, findings “not supported by source data”) → **June 2023**, the Svea Court of Appeal convicts him of gross assault against three patients, 2 years 6 months in prison.

Why it failed — five costumes, one failure

1. **No revascularization.** The trachea is supplied by many small segmental branches off the inferior thyroid artery, not by one named vessel. There is no pedicle to anastomose. Most organs come with a plug; a trachea comes with a hundred tiny wires you cannot reconnect. The graft has to wait for vessels to invade from surrounding tissue — weeks — and the tissue dies in days.
2. **No re-epithelialization.** The lumen needs ciliated respiratory epithelium to clear mucus. It never formed, and could not, because epithelium needs a vascularized submucosa beneath it.
3. **Infection and biofilm.** The airway is open to the outside. An avascular graft cannot defend itself — no blood means no immune cells arriving and **no antibiotic delivery**. You cannot treat an infection in tissue you cannot perfuse.
4. **Mechanical failure.** A trachea must stay rigid against negative pressure on every breath while flexing with the neck. Constructs collapsed and stenosed.
5. **Anastomotic breakdown.** Dead tissue at the join produced fistulas into adjacent structures. Several deaths came from this rather than the graft itself.

Two through five are all downstream of one. No blood supply means no epithelium, no immune defence, no antibiotics, and necrosis at the join. It is a single failure wearing five costumes — and it is the same failure as the necrotic core on the first slide, at twenty centimetres instead of two hundred microns.

Technical source if anyone wants it: Wei et al., “Tissue-engineered tracheal implants: Advancements, challenges, and clinical considerations”, *Bioengineering & Translational Medicine* 2024 (open access, PMC11256149).

Delivery note

There is a lurid subplot — fabricated CV, a relationship with a television producer, a Netflix documentary. **Do not open that door.** It will take five minutes you do not have and it turns a point about evidence standards into gossip. Keep it to: implanted before the biology was ready, results reported too positively, patients died, papers retracted, convicted.

11 Slide: Synthetic embryos

On screen. Figure 5C from [Fu et al. \(2025\)](#) — three whole mouse embryo models stained for OTX2/FOXP1, DAPI/SOX2/BRY and TX2/MYH2/GATA4.

Last rung. These are mouse embryos — except no egg was fertilized and no sperm was involved. They are built from stem cells alone. You take one line of stem cells, split it in three, and force two of the batches into the lineages that would normally become placenta and yolk sac by switching on a single transcription factor in each. Then you mix them together and let them self-organize. And they do: by day eight and a half you have a neural tube, a gut tube, and a beating heart. This is self-organization pushed as far as it goes — no scaffold, no printer, no blueprint, nothing but cells and the right environment. And it still stops, in a way that should now sound familiar. It only works inside a machine that supplies the oxygen. Even a natural mouse embryo in the same machine only reaches day eleven, and then dies — because there is no placenta, and nothing to plug its circulation into.

How you make an embryo out of only stem cells

An embryo needs three founding lineages: **epiblast** (the body), **trophectoderm** (placenta) and **primitive endoderm** (yolk sac). A fertilized egg makes all three itself. Here you force it: take three batches of the same naive ESCs and push two of them into extraembryonic identity by overexpressing **Cdx2** (trophectoderm) and **Gata4** (primitive endoderm), then aggregate. That is the ETiX/EiTiX system — the funnel at the left of the review’s Figure 5A.

If asked: why not just fertilize an egg in a dish?

You can. That is **IVF**, and it is routine — but it only works for about five or six days, to the blastocyst stage, after which the embryo must be transferred to a uterus.

Why five days: a human blastocyst is about **150–200 μm** across. It lives entirely on diffusion and its own stores, and at that size it does not need anything else. To grow past it, it must *implant* — burrow into the uterine wall and tap the maternal blood supply. IVF works right up to the diffusion limit and stops there.

Human embryos in attached culture have reached about 12–13 days (Shahbazi and Deglincerti, both 2016), which also runs into the **14-day rule**, so the technical and legal ceilings sit awkwardly close. Do not conflate them if asked.

Why the models are useful anyway: eggs are scarce and ethically loaded; stem cells are unlimited. You can make thousands of genetically identical models, so embryology becomes an *experiment* with n and controls rather than one shot at a unique genome. And you can watch and perturb the whole process in a dish, which you cannot do inside a uterus. This is also why they sit in an unsettled legal space — they did not arise from fertilization, and the ISSCR revised its guidelines specifically because of that.

The ex utero machine — what it actually takes

From [Aguilera-Castrejon et al. \(2021\)](#): natural mouse embryos grown from E5.5 to **E11** (hindlimb formation). What it required:

- Roller culture — rotating bottles on a drum, with electronic gas and pressure regulation.
- Medium: 25% DMEM, 50% rat serum, 25% human umbilical cord blood serum, plus extra glucose every 24 h.
- **A ramped oxygen schedule:** 5% O_2 at E7.5, 13% at E8.5, 18% at E9.5, 21% at E10.5.
- **Gas pressure held at 6–7 psi**, above atmospheric, explicitly to enhance oxygen delivery.

This yielded $\sim 77\%$ normal development over four days. Then, in their words: “the embryos started to show abnormalities, yolk sac circulation abruption and pericardial effusion, and quickly died overnight. . . consistent with the development of hydrops fetalis due to **insufficient oxygenation and nutrient supply by the ex utero system (given the lack of maternal blood supply)**”.

They were already running raised pressure to force more oxygen in, and it still was not enough.

Why the synthetic models stop even earlier

Models reach **E8.5**; natural embryos in the same apparatus reach **E11**. So the synthetic ones fail about two and a half days sooner — limited by the mass-transport ceiling *and* by their own imperfection: low efficiency, frequently abnormal morphology, wrong lineage proportions, sloppy extraembryonic compartments. Be straight about this if asked: they are **models**, not embryos, and **no synthetic mammalian embryo model has ever produced a live animal**.

If asked: what about artificial placentas / artificial wombs?

Those exist, but for a completely different problem. The Partridge 2017 “biobag” lamb, CHOP’s EXTEND, the EVE system — all target **extremely premature infants**. The method: cannulate the fetus’s **umbilical vessels**, run blood through a *pumpless* oxygenator driven by the fetus’s own heart, float the whole thing in synthetic amniotic fluid. Preterm lambs have been maintained for up to about four weeks.

Four reasons none of it reaches an embryo model:

1. **It needs an umbilical cord to cannulate.** A 23-week fetus has vessels big enough. An E8.5 model is a couple of millimetres with nothing to connect — the trachea’s “no artery to sew onto” problem, three orders of magnitude down.
2. **It needs a working fetal heart**, because the circuit is pumpless by design.
3. **It only replaces gas exchange.** A placenta also does hormone synthesis, immune tolerance, selective transport and waste clearance.
4. **It starts where ex utero culture ends, with an enormous gap.** From below: E8.5 (models), E11 (natural mouse). From above: roughly human week 23. Everything between — all of organogenesis and most of growth — is unclaimed.

The biobag is **supportive**: it sustains a fetus that already exists and is already plumbed. Embryo models are **generative**: they build something from nothing but cells. Neither can become the other.

(Partridge details are from memory — the PDF in [papers/](#) labelled [biobag_lamb.pdf](#) is actually a duplicate of the ex utero paper. Re-download before quoting exact gestational days.)

The deepest reason this rung ends here

The Cdx2/Gata4 trick means synthetic embryos *do* build trophoctoderm and primitive endoderm — the precursors of placenta and yolk sac. So why can they not just grow their own placenta and become self-sufficient?

Because a placenta is a two-party organ. It is an interface between fetal tissue and *maternal* blood — maternal spiral arteries, a maternal circulation, an entire second body on the other side. Stem cells can build the fetal half. The maternal half is somebody’s uterus.

So on this rung the bottleneck genuinely changes. Everywhere below, biology’s answer to mass transport was a better vascular tree. Here its answer was: use somebody else’s.

12 Closing

So: a cerebral organoid dies in the middle at a couple of hundred microns. A printed tissue block dies in twelve hours without a channel through it. A trachea killed people because there was no artery to connect it to. And a mouse embryo dies on day eleven because there is no placenta. Every rung of this ladder is the same problem wearing a different costume — getting oxygen to the middle of something. Biology solved it once, with a vascular tree that goes down to ten microns and, at the very top, with the trick of borrowing somebody else’s bloodstream. We can print down to about a hundred and fifty microns, we can steal a tree from a dead heart and not manage to fill it, and we cannot borrow a bloodstream at all. That is the field.

If you want a lighter last beat before the group picture: the whole talk is one number. Two hundred microns — about three sheets of paper. Everything above that rung is people trying to get around it, and mostly not succeeding yet.

13 Accuracy traps — read this on the morning

Things that are easy to say and wrong:

- **Do not say flow is required for angiogenesis.** Sprouting is driven by *hypoxia*, via HIF → VEGF, and happens perfectly well without flow. That is exactly why it is a problem: vessels do form in organoids, and still fail. Flow is required for *remodelling, maturation and survival* — unperfused vessels regress.
- **Do not call the magenta region necrotic.** It is hypoxic — alive and starving. The necrotic region is the part that does *not* stain.
- **Do not quote 0.8 mm as “the diffusion limit.”** That number comes from the SWIFT viability figure and is a *death* front measured by calcein at 12 hours, with medium at 21% O₂ at the boundary. It is not the same quantity as the 200 μm penetration depth, and the two will not reconcile if someone measures the scale bar.
- **Do not claim the day-140 organoid looks bigger.** It does not; the panels are at different magnifications. Say the number instead.
- **Do not generalize the blood-brain-barrier material.** LaMontagne et al. (2022) is cortex-specific. The strategies and the failure modes generalize; the BBB and neurovascular-unit specifics do not carry over to liver, heart or kidney.
- **The ex utero culture is NOT run at 95% O₂.** It is a ramp — 5% at E7.5, 13% at E8.5, 18% at E9.5, 21% at E10.5 — plus 6–7 psi of gas pressure. (95% O₂ is the *SWIFT* perfusion number; do not merge them.)
- **Do not say the embryo “stops at the 200 μm limit”.** An E8.5 embryo is millimetres across and has its own beating heart and yolk-sac circulation. It is not diffusion-limited in the organoid sense — it dies because that circulation has nothing to plug into. Same class of problem, one level up.
- **The biobag is supportive, not generative.** It sustains an existing, already-plumbed fetus. It cannot grow anything from small, and it cannot help an embryo model.
- **Ott 2008 is rat, not human, and not mouse.** Rat hearts, neonatal *rat* cardiac cells, *rat* aortic endothelial cells. No human cells and no iPSCs. “Human” appears only as the 16-week-fetal yardstick.
- **Do not say a trachea “has no artery”.** It has no *single* artery — blood arrives through many small branches. The flat version is wrong and a clinician will catch it.
- **Do not conflate the two trachea papers.** 2008 was a decellularized donor trachea; 2011 was synthetic plastic. Different objects, different retraction dates.
- **Do not say “bioprinting resolution is 150–350 μm”** as a blanket claim. That is *extrusion*. Light-based methods go far finer — Grigoryan’s alveolus is photopolymerized. Say it as a trade-off (cell-dense inks extrude coarse; fine methods need dilute inks or thin layers), not as a single number for the whole field.
- **“Resolution” here means minimum feature size**, not positional precision — the gantry positions far more finely than it can deposit. For sacrificial-ink vasculature the channel is the negative of the filament, so feature size is the smallest lumen you can make. Fair to call it resolution; just know the distinction if pressed.

- **On the SWIFT hyperoxia panel:** 95% vs 21% O₂ is a comparison between two *perfused* tissues. The honest sentence is “once you have perfusion, more oxygen buys nothing” — not “hyperoxia cannot rescue an avascular tissue”, which that panel does not test.

References

- Aguilera-Castrejon, A., Oldak, B., Shani, T., Ghanem, N., Itzkovich, C., Slomovich, S., Tarazi, S., Bayerl, J., Chugaeva, V., Ayyash, M., Ashoukhi, S., Sheban, D., Livnat, N., Lasman, L., Viukov, S., Zerbib, M., Addadi, Y., Rais, Y., Cheng, S., Stelzer, Y., Keren-Shaul, H., Shlomo, R., Massarwa, R., Novershtern, N., Maza, I., and Hanna, J. H. (2021). Ex utero mouse embryogenesis from pre-gastrulation to late organogenesis. *Nature*, 593(7857):119–124.
- Atala, A., Bauer, S. B., Soker, S., Yoo, J. J., and Retik, A. B. (2006). Tissue-engineered autologous bladders for patients needing cystoplasty. *The Lancet*, 367(9518):1241–1246.
- Fu, Y., Fan, Q., Wu, Y., and Bao, M. (2025). Unlocking the potential of stem-cell-derived ‘synthetic’ embryo models. *Trends in Biotechnology*, 43(8):1849–1863.
- Idnani, R. and Khan, S. (2026). Engineering the future of organ transplantation: A comprehensive review of 3D bioprinting advances for organ bioengineering. *Bioprinting*, 53:e00463.
- LaMontagne, E., Muotri, A. R., and Engler, A. J. (2022). Recent advancements and future requirements in vascularization of cortical organoids. *Frontiers in Bioengineering and Biotechnology*, 10:1048731.
- Lancaster, M. A. and Knoblich, J. A. (2014). Organogenesis in a dish: Modeling development and disease using organoid technologies. *Science*, 345(6194):1247125.
- Ott, H. C., Matthiesen, T. S., Goh, S.-K., Black, L. D., Kren, S. M., Netoff, T. I., and Taylor, D. A. (2008). Perfusion-decellularized matrix: Using nature’s platform to engineer a bioartificial heart. *Nature Medicine*, 14(2):213–221.
- Qian, X., Su, Y., Adam, C. D., Deutschmann, A. U., Pather, S. R., Goldberg, E. M., Su, K., Li, S., Lu, L., Jacob, F., Nguyen, P. T., Huh, S., Hoke, A., Swinford-Jackson, S. E., Wen, Z., Gu, X., Pierce, R. C., Wu, H., Briand, L. A., Chen, H. I., Wolf, J. A., Song, H., and Ming, G.-l. (2020). Sliced Human Cortical Organoids for Modeling Distinct Cortical Layer Formation. *Cell Stem Cell*, 26(5):766–781.e9.